

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

BOA202-R

DevOps in practice for beginners

Derek Bingham

Senior Developer Advocate, ANZ
Amazon Web Services

Donnie Prakoso

Principal Developer Advocate, ASEAN
Amazon Web Services



Objectives

For you

Understand the DevOps principles, methodologies, and tools

Gain practical knowledge on IaC, CI/CD, continuous configuration, and observability

Learn how to start your own personal DevOps culture

For us

Provide you with a good experience during the Chalk Talk

Get direct feedback to improve AWS developer tools and services

To share the  about DevOps

Today, we are going to discuss...

***Infrastructure
as code***



***Continuous
configuration***



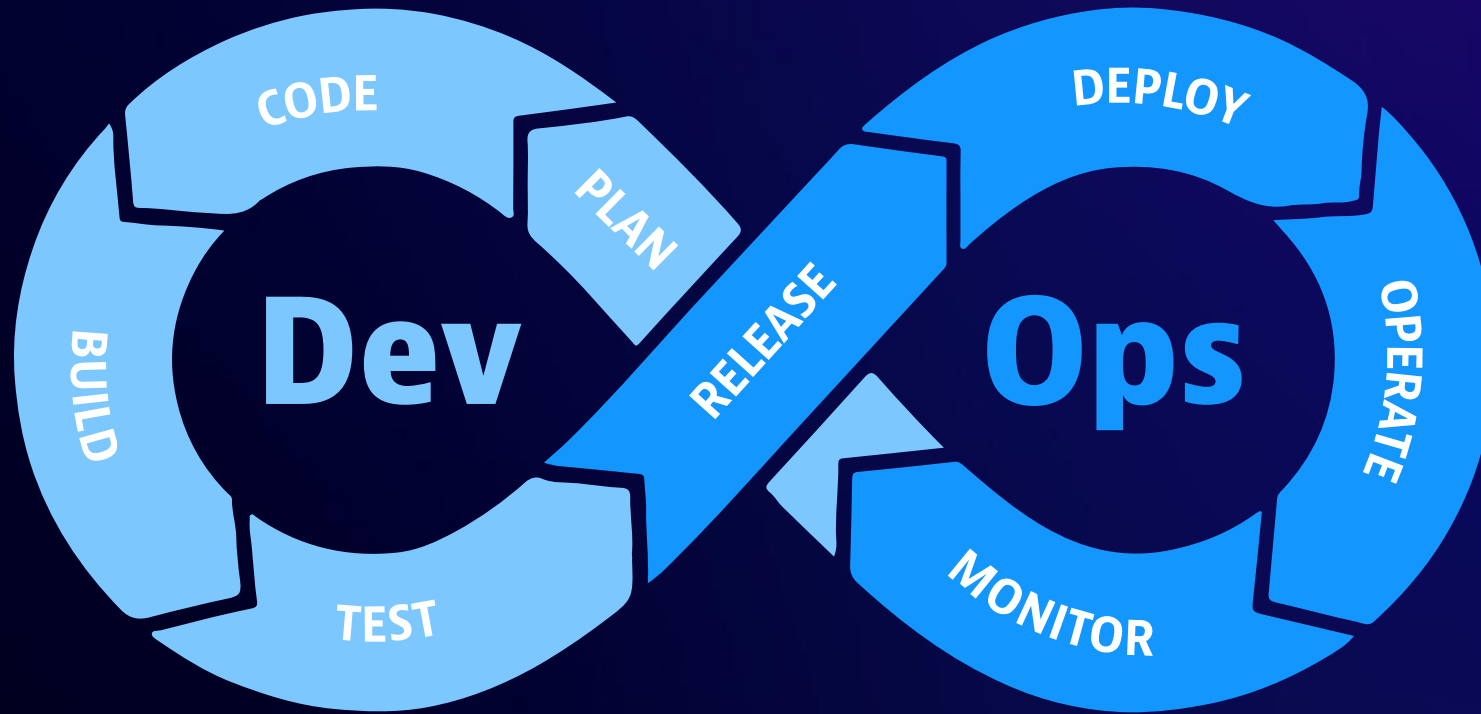
CI/CD



***Observability &
monitoring***



What is DevOps?



Teams that adopt modern software practices are more agile and higher performing

DEPLOYMENT
FREQUENCY

Weekly-monthly



Hourly-daily

CHANGE
LEAD TIME

1-6 months



1-7 days

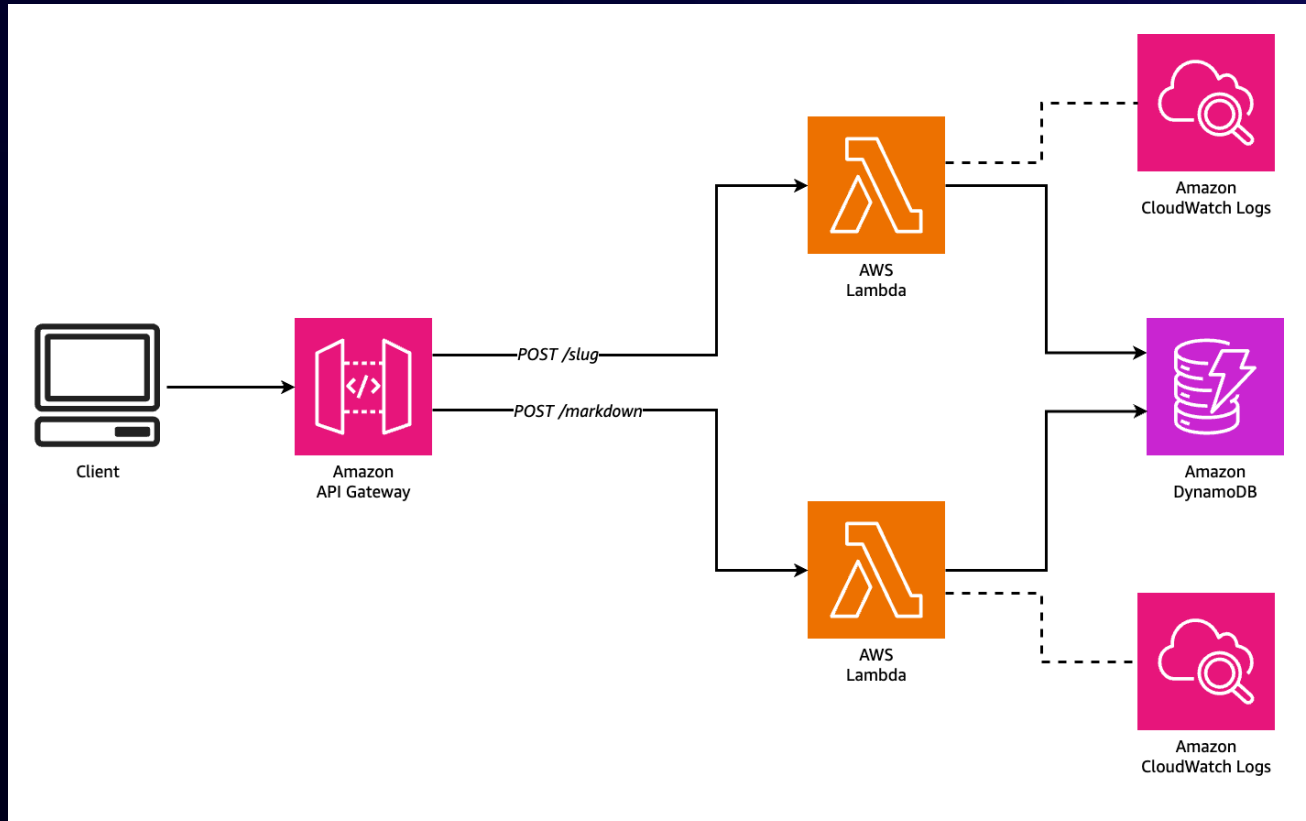
CHANGE
FAILURE RATE

46-60%



0%-15%

Let's start with a simple task



Build serverless APIs

Build a set of serverless APIs to process string operations from text input

We also need to save the request data into a database

Infrastructure as → is code

1. Make infrastructure changes repeatable and predictable
2. Release infrastructure changes using the same tools as code changes
3. Replicate production environments in staging environments to enable continuous testing

AWS Cloud Development Kit (CDK)

A MULTI-LANGUAGE SOFTWARE DEVELOPMENT FRAMEWORK FOR MODELING CLOUD INFRASTRUCTURE AS REUSABLE COMPONENTS

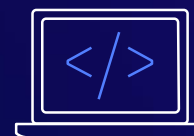
```
class UrlShortener extends cdk.Stack {
  constructor(scope: cdk.App, id: string, props?: cdk.StackProps) {
    super(scope, id, props);

    const vpc = new ec2.Vpc(this, 'MyVpc', { maxAzs: 2 });
    const cluster = new ecs.Cluster(this, 'Ec2Cluster', { vpc });
    cluster.addCapacity('DefaultAutoScalingGroup', {
      instanceType: ec2.InstanceType.of(ec2.InstanceClass.T2, ec2.InstanceSize.MICRO)
    });

    // Instantiate ECS Service with just cluster and image
    const ecsService = new ecs_patterns.NetworkLoadBalancedEc2Service(this, "Ec2Service", {
      cluster,
      memoryLimitMiB: 512,
      taskImageOptions: {
        image: ecs.ContainerImage.fromRegistry("amazon/amazon-ecs-sample"),
      }
    });

    ecsService.service.connections.allowFromAnyIpv4(EphemeralPortRange);

    new cdk.CfnOutput(this, "networkLoadBalancerURL", {
      value: "https://" + ecsService.loadBalancer.loadBalancerDnsName,
      description: "Network LoadBalancer URL"
    });
  }
}
```



Familiar
Your language – just code



Tool support
Autocomplete – inline
documentation



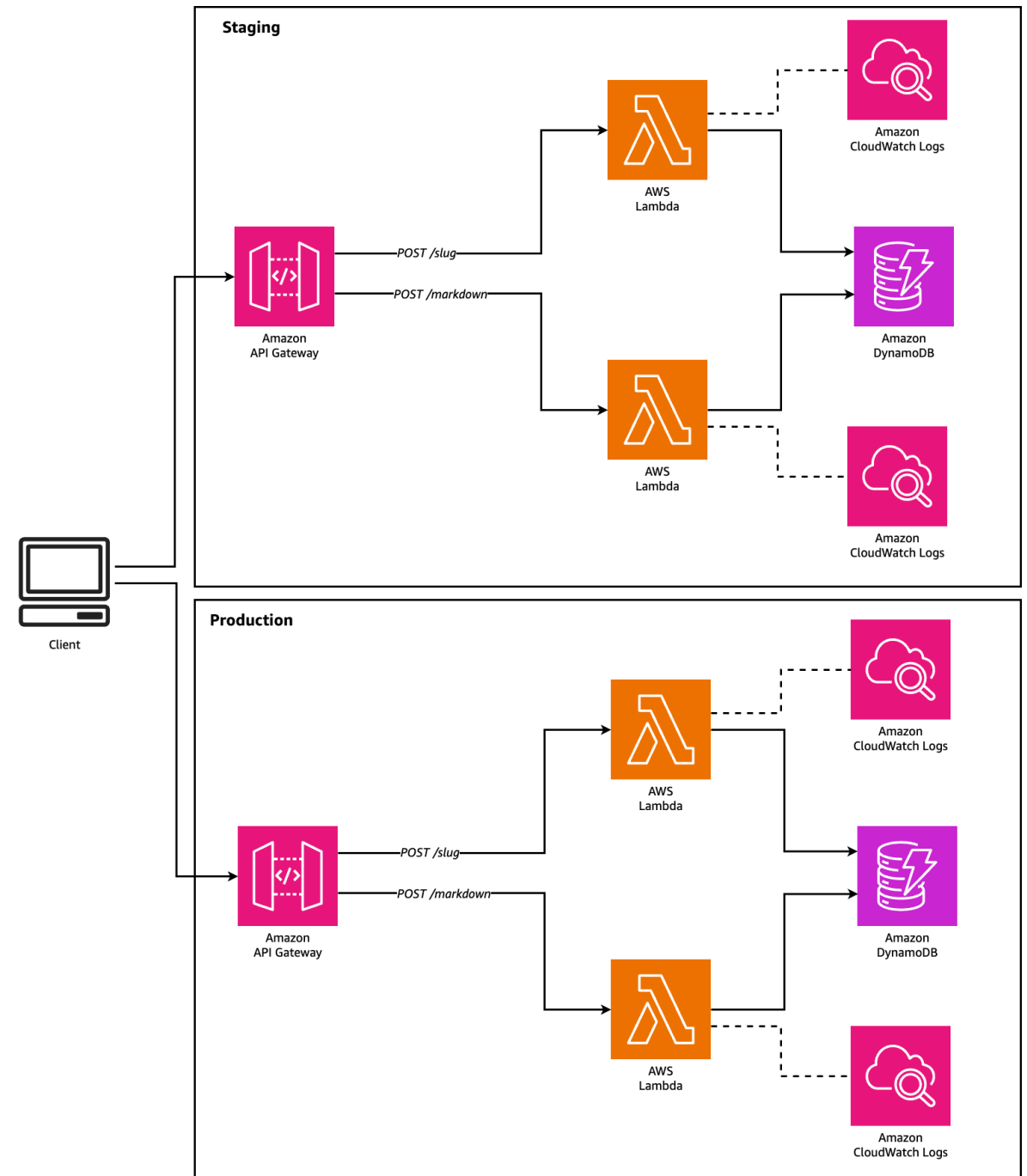
Abstraction
Sane defaults – reusable
classes



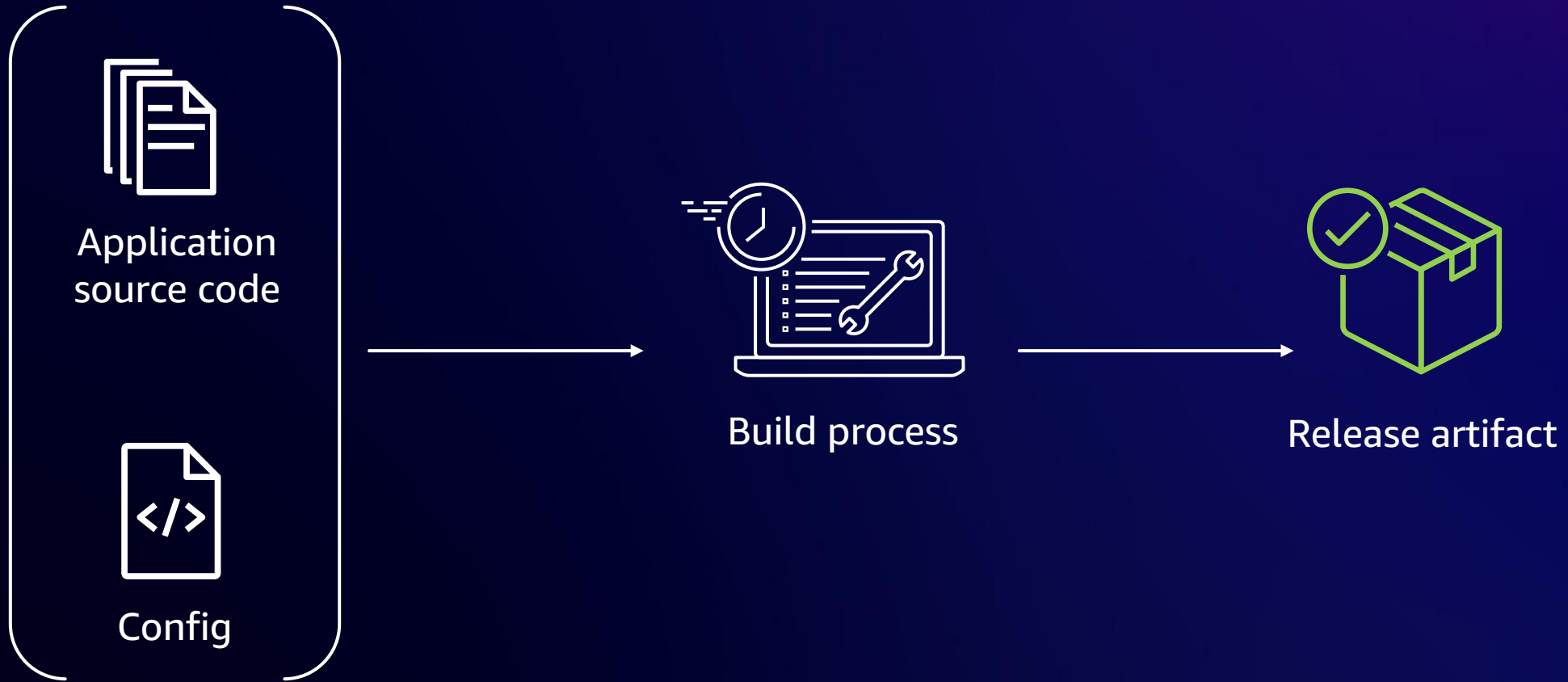
But we need to build multiple app environments

Configure multiple environments for previous serverless APIs

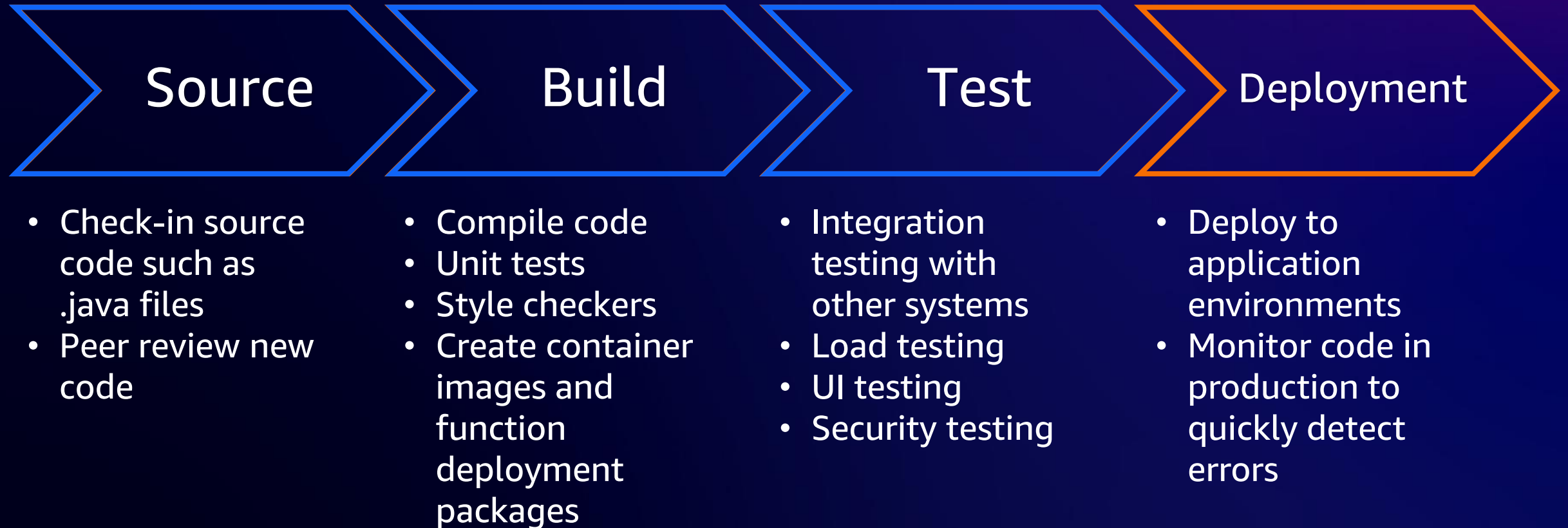
Deploy the same stack for environments: Maintain dev/prod parity



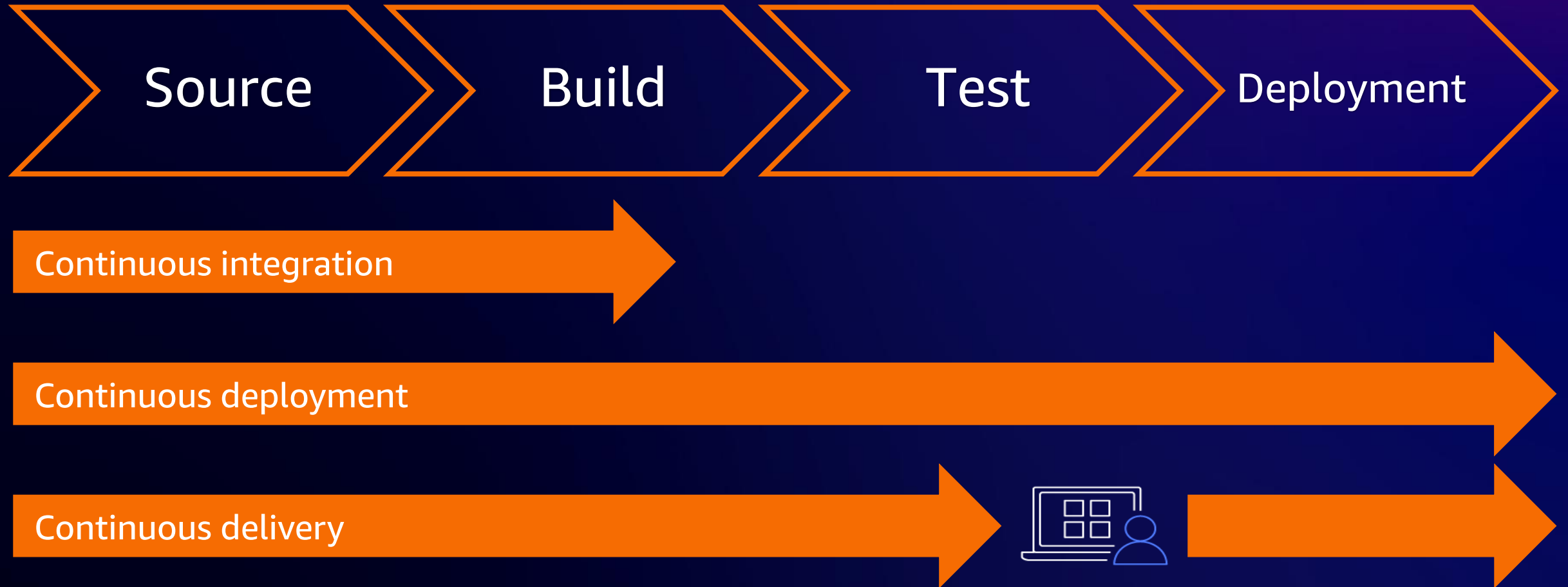
Decoupling configuration from applications



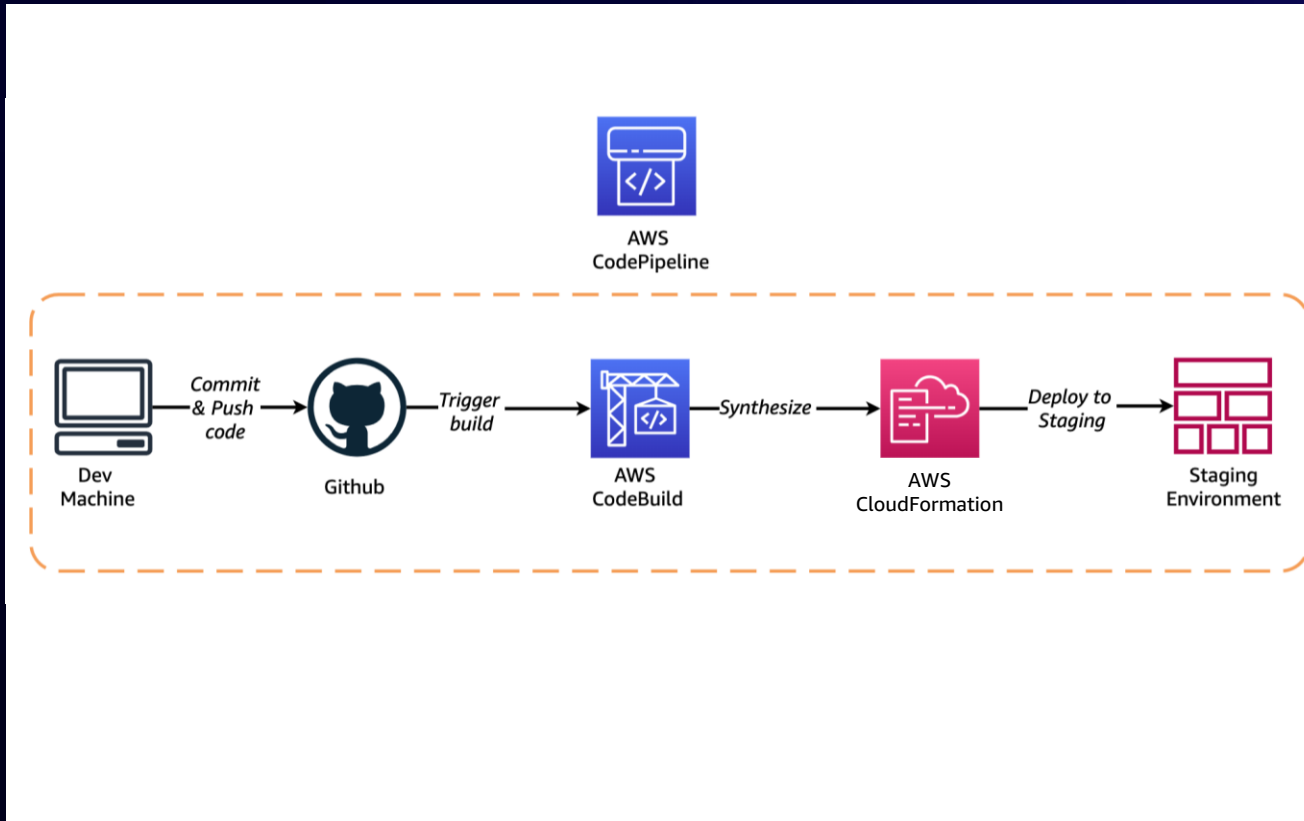
Release process stages



Release process stages



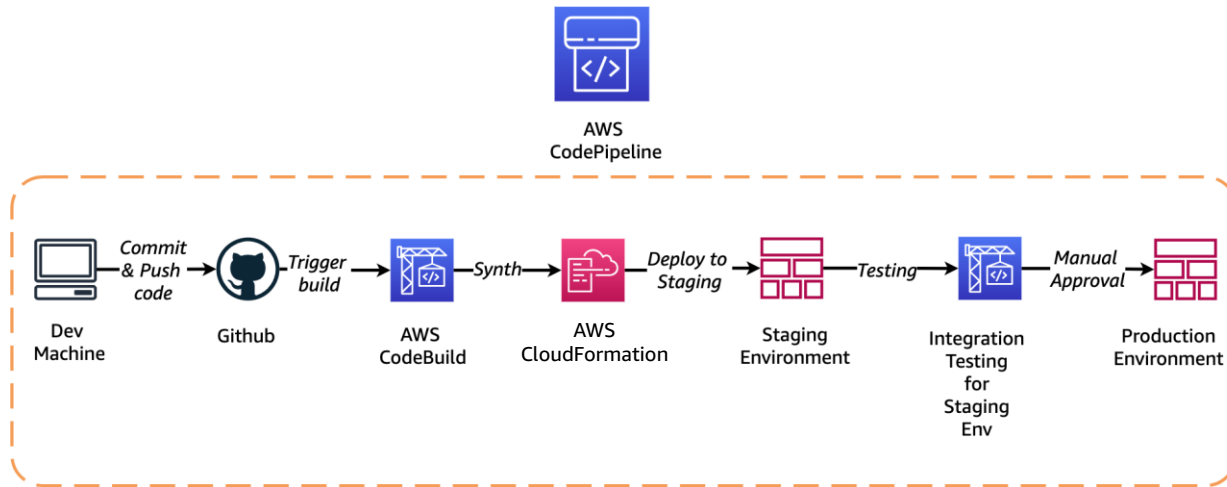
Implement CI/CD with CDK pipelines



Building release pipeline

Automate deployment when new code is pushed and creates self-mutating infrastructure in the application environment

Continuous Delivery for multi-environments



CI/CD for multi-environments

Create a release pipeline that self-mutates and is triggered when a new code is pushed

Implement continuous deployment in the staging environment and continuous delivery (with manual approval) for production

GitOps

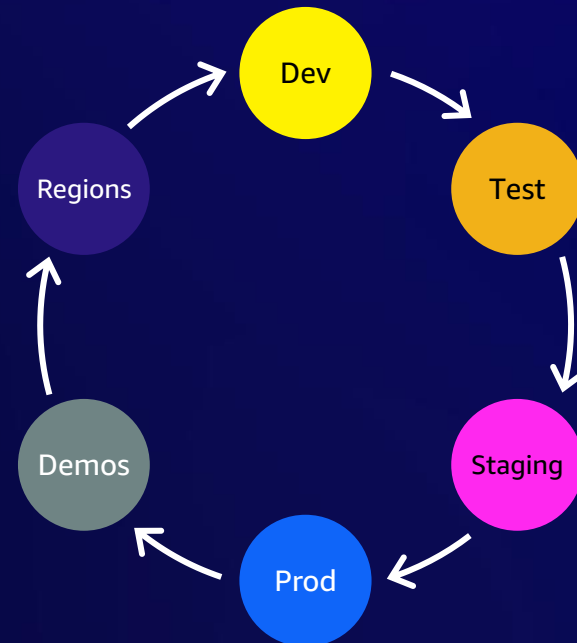
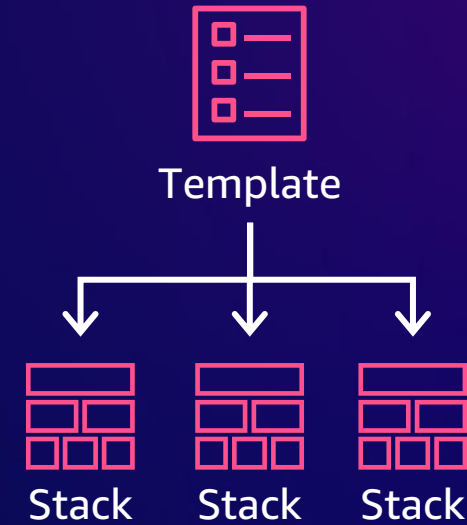
Single source of truth to deploy the whole stack

A single workflow to control versioning on your infrastructure

Service rolls back to the last good state on failures

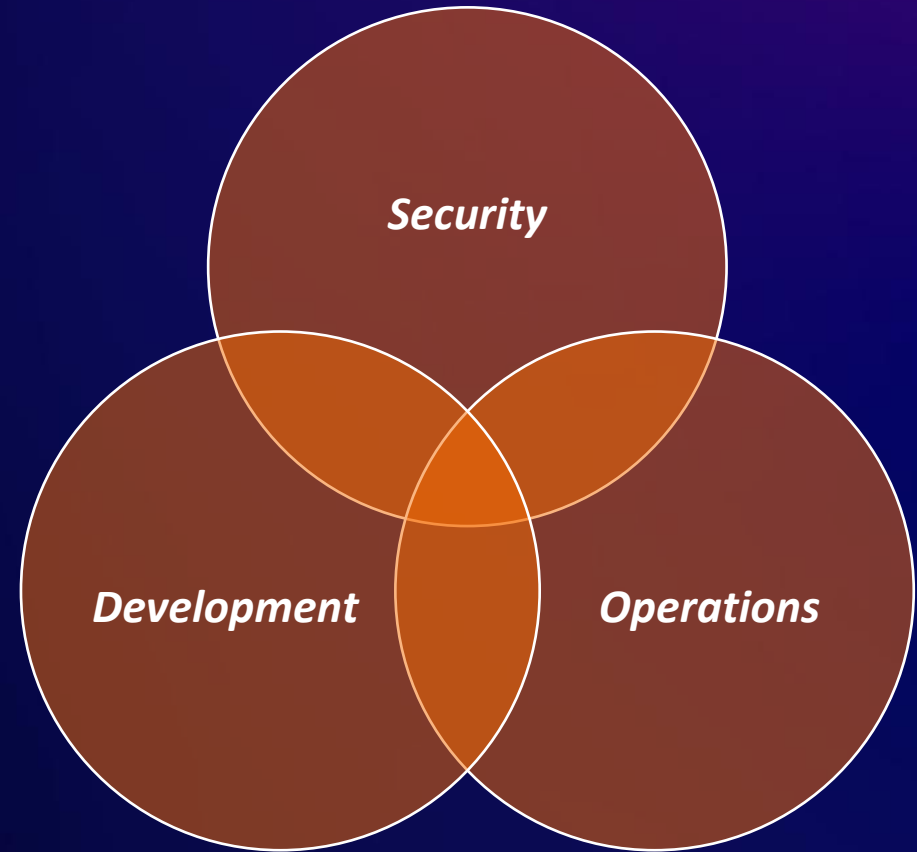
Infrastructure is defined, built, and run through CI/CD

Drift detection



What is DevSecOps?

DevSecOps is achieved by **integrating and automating** the enforcement of preventive, detective, and responsive security **controls** into the pipeline



Observability and monitoring: What's happening?

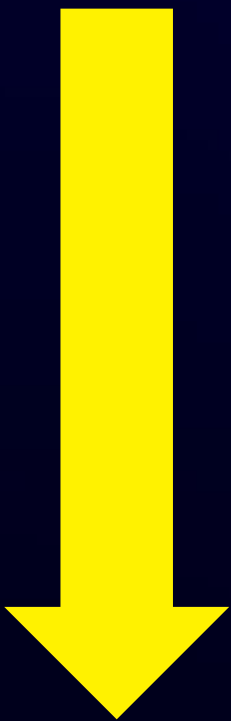
The “three pillars of observability”

“MeLT”



Observability stack

Outside-in



Digital experience

Application

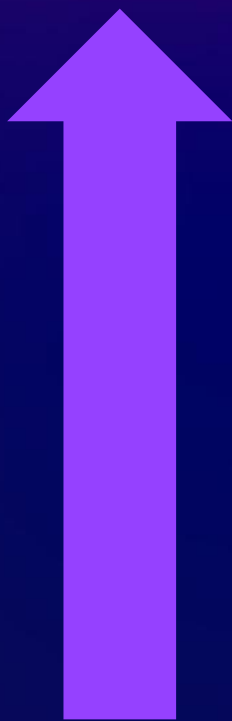
Infrastructure

AWS services

CloudWatch Synthetics
CloudWatch RUM
CloudWatch Evidently

AWS X-Ray insights
CloudWatch ServiceLens, CloudWatch Container Insights,
CloudWatch Lambda Insights, Contributor Insights,
CloudWatch Application Insights

CloudWatch Logs, alarms, metrics, and dashboards



Inside-out



DevOps – next steps



DevOps Workshop



Observability Workshop



CDK Workshop

Thank you!



Please complete the session
survey in the mobile app

Derek Bingham

Senior Developer Advocate, ANZ
AWS

Donnie Prakoso

Principal Developer Advocate, ASEAN
AWS

